

Drone-based radiotelemetry and imagery systems provide an advantage over traditional techniques for estimating survival of dependent juveniles

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Abstract

Traditional methods for estimating juvenile survival in wildlife populations often rely on marking and tracking individual young. However, traditional methods can introduce bias into survival estimates via the capture or handling process and are logistically challenging to implement. Alternatively, researchers studying avian and mammalian species that provide parental care can mark and monitor adults to estimate survival probability of the unmarked offspring. We implemented a field sampling approach using unoccupied aerial vehicles (UAVs, i.e., drones) to estimate detection probabilities and pre-fledging survival of blue-winged teal (*Spatula discors*) by marking and resighting adult females with broods ($n = 27$). We used a combination of 2 methods to detect marked females, drone-based VHF radiotelemetry and multispectral imagery, to estimate brood detection and apparent survival using Cormack-Jolly-Seber models. We compared between the VHF drone and the camera drone to investigate the ability of the VHF drone to locate marked females and thermal cameras to detect broods. Detections using the camera drone were informed by prior location cues from the VHF drone, meaning that detections were not independent but instead

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relied on the cues provided by the VHF drone. We monitored marked females and their unmarked broods weekly during the summer of 2023 in Saskatchewan, Canada. Of marked females, 9 suffered complete brood loss (33%), 4 had unknown fates (15%), 14 had at least one duckling survive to 35 days (52%), and we observed that 120 of 230 ducklings survived. Detection probability for sampling with the VHF drone remained constant over time (mean = 0.920), whereas detection probability using the camera drone increased with each sampling occasion (mean = 0.756). Weekly brood survival probability increased with brood age and were similar between survey methods (VHF: mean = 0.907; camera: mean = 0.898). Capitalizing on the interdependence of unmarked juveniles and marked adults, and drone-based methodology, our approach allowed us to effectively and efficiently estimate juvenile survival for waterfowl. Our approach is applicable to a range of other species as well, where providing more precise and efficient field sampling methods for monitoring vital rates and population dynamics can enhance ecological understanding and resultant management practices.

KEYWORDS

broods, cameras, drones, duckling, *Spatula discors*, survival, telemetry, waterfowl

Understanding how vital rates influence population dynamics is critical to effectively manage wildlife, and variability in reproduction plays a dominant role in driving population trajectories for many species (Hoekman et al. 2002, Clark and Martin 2007, Taylor et al. 2012). Estimating the proportion of offspring produced that survive to join the breeding population (i.e., recruitment) has been a central focus of ecological research (Mayfield 1961, Street et al. 2022). For many bird species, recruitment can be subdivided into stages such as nest survival (i.e., probability a nest survives to hatch ≥ 1 egg), juvenile survival from hatch to fledging, and post-fledging survival to the following breeding season, each of which represents a discrete biological metric with inherently different survival probabilities. While methods for studying nest survival are relatively well established, those for studying juvenile survival are challenging to implement and usually requires marked juveniles that are tracked or reencountered. As such, methods for studying juvenile survival are logistically challenging to implement and can introduce bias into survival estimates. (Pietz et al. 2003, Nicolai and Sedinger 2012, Sedinger et al. 2018). Marking young individuals can negatively impact survival either in the handling process or via recapture (Amundson and Arnold 2010, Barron et al. 2010, Davis et al. 2016). Moreover, the challenges of capturing juveniles which attain mobility quickly after hatch highlight the need for non-invasive and more efficient methods for estimating juvenile survival (Amundson and Arnold 2010, Riley and Conway 2020).

Pre-fledging juvenile survival has been particularly difficult to estimate for avian species that have precocial offspring that are cryptic and elusive (Pagano and Arnold 2009, Colwell 2010, Riley et al. 2021). For example, in waterfowl, time-activity budgets produced by Ringelman and Flake (1980) documented that duck broods (i.e., attending female and pre-fledged offspring) were not visible to observers. Accordingly, pre-fledging survival is relatively understudied and difficult to estimate compared to nest survival (Klett et al. 1988, Greenwood et al. 1995), primarily due to the inability to

effectively detect broods concealed in dense vegetation (Cowardin et al. 1983, Hoekman et al. 2004). Compounding challenges in detection are tracking brood movements during the pre-fledging period as most broods relocate several times (Rotella and Ratti 1992, Krapu et al. 2006). Estimating pre-fledging survival is thus hampered by two primary limitations: 1) offspring are too small to capture and mark without affecting their survival (Amundson and Arnold 2010, Barron et al. 2010, Davis et al. 2016) and 2) brood behavior and movements reduce their detectability (Diem and Lu 1960, Rotella and Ratti 1992, Pietz et al. 2003). An alternative approach to marking offspring is to mark adults with dependent offspring, as in waterfowl, and use repeated observations of the adult and offspring to estimate juvenile survival (Brownie and Robson 1983, Orthmeyer and Ball 1990, Lukacs et al. 2004).

Adults are easier to capture and mark than offspring, but traditional juvenile survival studies still suffer from a number of pitfalls. Ground-based radio-tracking techniques are limited to tracking a limited number of individuals at a time, which is labor-intensive when applied over large study areas (Mech 1983). In addition, signals are often attenuated by terrain features, such as valleys, ridges, or dense forest cover (Mech 1983). As a result, researchers often need to seek higher ground to increase the likelihood of being within the line of sight of transmitters and detecting signals (Mech and Barber 2002). Fixed-wing aerial surveys have been used to find missing individuals, but aerial surveys provide general locations, requiring a second ground-based trip to ensure accuracy (Grand and Flint 1996, Yetter et al. 2018). Although telemetry can confirm the presence of a marked individual, ground-based visual methods are required to obtain counts; but reliable, accurate counts without causing disturbance are difficult (Bloom et al. 2012, Riley and Conway 2020). The lack of suitable alternatives to estimate juvenile survival has likely contributed to the common assumption of perfect detection from counts from unmarked offspring groups (Flint et al. 1995, Manly and Schmutz 2001), which can introduce bias into survival estimates.

Unoccupied aerial vehicles (hereafter UAVs or drones) are increasingly being used to monitor wildlife populations (Chabot and Bird 2015, Linchant et al. 2015, Hodgson et al. 2016). Drones enable observation of wildlife across extensive areas with greater efficiency, less disturbance, and more precision compared to ground-based and fixed-wing aircraft observations (Chabot and Bird 2012, Hodgson et al. 2018, Pöysä et al. 2018). Furthermore, high-resolution optical and thermal imaging systems significantly enhance the detection and identification of cryptic species with elusive life history traits (Cilulko et al. 2013, Scholten et al. 2019, Spaan et al. 2019), inclusive of juvenile waterfowl (Bushaw et al. 2021, Mitchell et al. 2023). Recording imagery associated with observations creates a permanent record, providing the opportunity to error-check or recount observations made in the field, unlike ground-based methods. The recent integration of radiotelemetry systems into drones has improved our ability to track a diversity of species over greater survey areas compared to ground-based radiotelemetry (Cliff et al. 2018, Saunders et al. 2022). Additionally, using a radiotelemetry system mounted on a drone overcomes many of the signal-based challenges experienced when tracking with ground-based radiotelemetry (Cliff et al. 2018). When combined with multi-spectral imaging cameras that can improve detection rates, drone-based radiotelemetry is a powerful system for relocating and monitoring mobile and elusive animals.

Here, we present a field sampling approach to monitor pre-fledging juvenile survival in blue-winged teal (*Spatula discors*) broods, using drones equipped with a real-time radiotelemetry system and multi-spectral thermal and optical cameras. We implemented a hierarchical marking system of adult females that can be used to estimate and compare detection and survival probabilities across different monitoring protocols. Our primary purpose here is to describe the field methods and equipment, and demonstrate how they might be applicable for studying juvenile survival across a suite of animal species.

STUDY AREA

We collected data from radio-marked duck broods on 5, 64.75 km² study sites located between Humboldt and Saskatoon, Saskatchewan (52°12' N, 105°07' W, 52°07' N, 106°39' W, respectively) from June to August of 2023. The study sites had similar mean monthly precipitation (118–136 mm) and minimum and maximum temperatures

(12–25°C; Climate Engine; Huntington et al. 2017). All study sites were characterized by moderate densities of seasonal wetlands, and high densities of semi-permanent and permanent wetlands (Stewart and Kantrud 1971). Seasonal and semi-permanent wetlands were characterized by open water and bands of emergent vegetation surrounding the shoreline, while permanent wetlands tended to have exposed shoreline. The landscape was highly fragmented, consisting primarily of cropland (cereal grain and oil-seed production) and introduced and native grassland cover used for managed grassland, hayfields, and pasture for cattle. Our research was conducted on idle grasslands and cattle pastures enrolled in easements or fee-titled acquisitions through Ducks Unlimited Canada, as well as privately owned fields used for hay land.

METHODS

We searched for blue-winged teal nests during the 2023 nesting season from mid-May to early July. Nests were found by dragging a chain between two all-terrain vehicles to flush females from nests (Klett 1986). We then candled eggs to determine the developmental stage of the eggs to estimate a hatch date (Weller 1956). We constructed 5-m diameter welded-wire predator exclosures around nests at their initial discovery to increase the probability a nest would survive to hatch (Sargeant et al. 1974).

To reduce nest abandonment, we waited until the nest was late in incubation (>15 days) before attempting to capture females using dip nets and nest traps (Weller 1957). All captured females received an aluminum leg band (U.S. Geological Survey; size 4a) and were fitted with a 4-g prong-and-suture radiotransmitter equipped with a mortality sensor (VHF; Advanced Telemetry Systems, Isanti, MN, USA; Figure 1). Additionally, females were fitted with nasal saddles that were hand-crafted from polyvinyl chloride, shaped to fit the contours of the duck's bill using a molded blue-winged teal replica, and engraved with a symbol and a character to denote the study area and the individual female (Sugden and Poston 1968, Cook et al. 2021). In the field, we affixed the saddle using an aluminum rod passed through the duck's nares, secured with a set of stainless-steel washers and by crimping the aluminum rod. For the prong and suture implant, we identified the best location mid-dorsally, just anterior to the shoulder joint of each female (Pietz et al. 1995). We then used iodine to clean the site and applied 2, 1.5-ml injections of Lidocaine before using a small cannula to make a small incision where the prong was inserted. Finally, we used a single suture to attach the posterior end of the transmitter to the skin. All surgery equipment, including the transmitter, were immersed in a sterilizing solution prior to each installation, and field personnel wore appropriate personal protective equipment.



FIGURE 1 A blue-winged teal female marked with an aluminum leg band, a uniquely coded nasal saddle denoted by a symbol and a character (x0), and a 4-g prong-and-suture radiotransmitter.



FIGURE 2 From left to right: the VHF drone (DJI® Matrice 210 UAV quadcopter equipped with the radio-receiver payload from Wildlife Drones®) and the camera drone (DJI® Matrice 30 T UAV quadcopter equipped with integrated thermal and optical cameras).

We located radio-marked females using a battery-operated DJI® Matrice 210 quadcopter drone (6.4-kg weight, 716 × 220 × 236 mm dimensions; Da-Jiang Innovations, Shenzhen, Guangdong, China), powered by 2, 22.8 V lithium pro ion batteries that permitted surveys of ~30 minutes before changing batteries. The drone was equipped with a radio-receiver payload and was associated with a computer base station from Wildlife Drones® (Figure 2). The radio-receiver payload consisted of a VHF radio-tracking sensor and directional antenna, which simultaneously monitored up to 40 different transmitter frequencies and sent all radio-tracking data to the computer base station for processing and mapping in real time using proprietary software from Wildlife Drones® (hereafter VHF drone; Saunders et al. 2022).

We located broods and counted ducklings using a battery-operated DJI 30 T quadcopter drone (3.8-kg weight, 470 × 585 × 215 mm dimensions; Da-Jiang Innovations, Shenzhen, Guangdong, China), powered by 2, 26.1 V lithium pro ion batteries that permitted surveys of ~45 minutes before changing batteries. The drone was equipped with an integrated multispectral camera (wide, zoom, thermal, and laser rangefinder) that could be toggled using the flight controller tablet (Figure 2; hereafter camera drone). We used the thermal imaging camera (640 × 512 resolution; 9.1 mm lens; white-hot filter) to detect the thermal signature of waterfowl broods and then used the optical camera (3840 × 2160 resolution, 21--75 mm lens) to count ducklings in each detected brood. An inverter generator and portable battery charging stations were used to charge batteries in the field allowing us to operate continuously throughout the day. A desktop monitor was connected to the flight controllers and computer base station via an HDMI cable to enhance the observation process.

We monitored marked blue-winged teal females on the nest daily using the VHF drone to determine nest status and termination date. Observers worked in 2-person pilot and observer teams to track radio-marked females. The drone pilot launched the VHF drone >100 m from the nest and ascended to 61 m in elevation to increase the likelihood of detecting a signal and reduce disturbance to the nesting female (Mech and Barber 2002, Vas et al. 2015), flying at a speed of 0–6.7 m/s. Since the nest location was known, the VHF drone encircled the nest to acquire a strong signal, indicating female presence or absence. Signal detection was denoted on the computer base map as colored dots that represented signal pulses per minute (PPM) of the radiotransmitter: black (no signal), blue (2–19 PPM), green (20–70 PPM), and red (71–100 PPM; mortality). The closer the VHF drone is to the transmitter, the stronger and more consistent the pulses are received. If the female was not on the nest or had been killed, we then approached the nest on foot to determine whether the female was on an incubation recess, or if the nest had been destroyed or abandoned. If the nest was still active, we continued to monitor the nest on a daily basis until fate was determined. After hatching, we counted the number of egg membranes to determine the number of ducklings within the brood (Klett et al. 1986).

After the successful hatch of a nest, drone pilots launched the VHF drone >100 m from survey wetlands near the hatch location and ascended to an elevation of 61 m to locate the brood (Mech and Barber 2002, Vas et al. 2015). We conducted drone flights manually rather than by pre-determined flight plans because the display of real time direction and signal strength data overlaying satellite imagery basemaps enabled the drone pilot to adjust the flight path as needed for efficient data collection. Pilots started by searching all wetlands within the immediate vicinity of the attending female's nest site and expanded their search area until the radio signal was detected or we determined that the female left the survey area. Once a radio signal frequency was detected, we encircled each wetland in the area and used the signal strength filter to dynamically identify which wetland the female was using. As radio signal frequencies were detected with increasing strength, the drone pilot would decrease flight speed to ascertain the bird's most likely position in the wetland. As an alternative, triangulation could be used to determine the location of the radio-marked females; however, this approach required at least 3 full rotations of the drone at separate locations to obtain bearings to the strongest signal, upon which an estimated error ellipse was automatically generated around the azimuth intersections by the Wildlife Drones[®] software and displayed on the base map.

Using wetland location cues from the radio-marked female, we deployed the camera drone at an elevation of 46 m to locate broods, resight nasal saddles, and count ducklings using the thermal and optical cameras. The camera drone was deployed at a lower elevation than the VHF drone to increase the thermal camera's capability to detect the brood while still avoiding disturbance to the brood (Vas et al. 2015). Depending on size, wetlands could be surveyed by rotating the drone at a single observational point, and for others it was necessary to circumnavigate the entire wetland and all the emergent vegetation in modified transects (Bushaw et al. 2021). Because all operation was manual, flight speed of transects varied (0–4.5 m/s) as potential broods were detected. Once a brood was detected with the thermal camera, the operator switched to the optical camera and used the zoom function to identify the nasal saddle on the female, ensuring this was the same brood located with the VHF drone, and counted and aged the ducklings (Gollop and Marshall 1954; Figure 3). We used recorded images and video to confirm duckling counts associated with each female once we left the field.

Duckling mortality is highest during the first 2 weeks post-hatch (Ball et al. 1975, Talent et al. 1983, Gendron and Clark 2002), so we attempted to count ducklings within 3- and 7- days after hatch. Following the initial sightings of a brood, we attempted to resight all broods once per week. We monitored each brood until ducklings reached 35 days, the entire brood was lost, or the attending female left the brood as they grew older and more independent. Therefore, each brood was potentially observable 6 times: twice in the first week of brood-rearing, and weekly after that until the ducklings reached 35 days of age or the attending female left the brood, or all of the ducklings died. We did not monitor ducklings past 35 days of age because survival remains relatively constant thereafter (Krapu et al. 2006, Amundson and

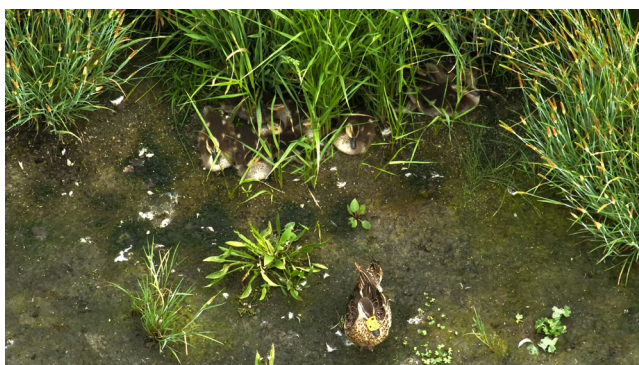


FIGURE 3 A radio-marked female, identified by nasal saddle code square J, and her associated brood in the emergent vegetation behind her (9 ducklings at age class 1c; Gollop and Marshall 1954).

Arnold 2011). Like many precocial species, behavior of the attending female changes when the entire brood is lost (Pietz et al. 2003). When females were repeatedly observed alone, or on wetlands associated with large congregations of non- and post-breeding waterfowl, we assumed complete brood loss and monitored females weekly to check for renesting. Attending females left their brood between 21 to 35 days after hatch; we assumed independent ducklings >20 days survived to fledging based on previous research (Gendron and Clark 2000).

We estimated brood detection (p) and apparent survival (Φ) probabilities using the Cormack-Jolly-Seber open capture-recapture model (hereafter CJS; Cormack 1964, Jolly 1965, Seber 1965), employing the state-space formation (Gimenez et al. 2007, Royle 2008, Kéry and Schaub 2011). We organized resight observations (y) in an encounter history matrix, with each row representing brood i and columns indicating resight occasion t . Our encounter histories consisted of entries that were either a 1 or 0, with a 1 indicating brood i was alive and resighted at occasion t ; 0 indicated brood i was not resighted at time t , indicating that it was either alive and not detected or dead. To estimate Φ , the latent state (z ; i.e., alive = 1, dead = 0) of brood i was defined each resight occasion t . The state process was defined as:

$$z_{i,t+1}|z_{i,t} \sim \text{Bernoulli}(z_{i,t}\Phi_{i,t}),$$

where the z of brood i at resight occasion $t + 1$ was conditional on being alive at resight occasion t . The CJS model considers the product of z and Φ in deriving the Bernoulli success parameter to ensure brood deaths ($z = 0$) occurring at time t have no influence on Φ estimates at time $t + 1$. The observation process used was defined as:

$$y_{i,t}|z_{i,t} \sim \text{Bernoulli}(z_{i,t}p_{i,t}),$$

where y of brood i at time t was conditional on the z of brood i at time t . The CJS model considers the product of z and p in deriving the Bernoulli success parameter to ensure brood deaths ($z = 0$) occurring at time t have no influence on p estimates at time $t + 1$. The time intervals for the first encounter occasion (3 days post-hatching) and the subsequent weekly encounters were unequal; therefore, we adjusted the survival estimate for the first encounter occasion by exponentiating it to the third power (White and Burnham 1999). We fit our CJS models within a Bayesian framework in JAGS (Plummer 2003), using the package jagsUI (Kellner 2019), in R version 4.2.3 (R Core Team 2023). We ran 4 chains of 10,000 iterations, with a burn-in period of 5,000 for each saved iteration. Model convergence was confirmed via visual inspection of trace plots and the $\hat{R}$ statistic (Gelman and Rubin 1992). We ran two models to estimate Φ and p for broods that were resighted with 1) the VHF drone and 2) the camera drone with location cues from the VHF drone to ascertain how reliable the VHF drone was for locating broods and evaluating the effectiveness of our thermal cameras for detecting broods with known locations. Results from our CJS models are presented as posterior means (μ) and 95% credible intervals (hereafter 95% CI).

RESULTS

We radio-marked 46 blue-winged teal females from late June to early August 2023. Of those 46 radio-marked females, 19 females had nests that failed to hatch, primarily due to predation ($n = 12$) and abandonment ($n = 3$). The remaining 27 radio-marked females had nests where ducklings hatched, of which mean brood size at hatch was 9 (SD = 0.11). Total brood loss was experienced by 9 females (33%), while 14 broods had at least 1 duckling survive, which were considered successful broods (52%; Dzuz and Clark 1998). Additionally, 4 broods had unknown fates (15%) due to transmitter failure or emigration from the study area. Of the broods that had known fates, we observed that 120 of 230 ducklings survived.

We found that circling individual wetlands to locate radio signal frequencies with the VHF drone was more efficient than triangulation techniques because our potential brood locations were inherently restricted to wetlands



FIGURE 4 The dynamic search method involves flying the VHF drone around individual wetlands to detect radio signals from radio-marked females. The path of the VHF drone is displayed against a satellite image basemap with recordings of signal pules per minute (PPM; black, blue, green, and red dots). The legend on the lower left shows the color-coded ranges of PPM.

(Figure 4). By filtering the radio signal frequency to its highest strength, we could precisely locate marked females (Figure 5). Triangulation search methods provided locations that varied in accuracy, dependent on the transmitter's proximity, and required a minimum of 30 minutes of battery life, necessitating battery changes before tracking subsequent individuals (Figure 6).

Our CJS models successfully converged according to trace plot inspections and $\hat{R}$ values. The mean brood detection probability was 0.920 (CI = 0.779–0.991; Figure 7) and apparent survival probability was 0.907 (CI = 0.743–0.989; Figure 8) using the VHF drone. Comparatively, mean brood detection probability was 0.756 (CI = 0.571–0.823; Figure 7) and apparent survival probability was 0.898 (CI = 0.720–0.993; Figure 8) using the camera drone, indicating that there were some broods detected with the VHF drone that were not detected visually, even after intensive searching. Note that the 2 detection probabilities are not completely independent, as the VHF drone was used to guide the camera drone flights, but highlights that thermal cameras do not detect the presence of all broods.

DISCUSSION

We developed a field sampling technique to estimate survival of unmarked juveniles using drone-based technology and marked adults, while accounting for imperfect detection. The integration of drones equipped with real-time radio-tracking technology and multi-spectral thermal and optical cameras can address some of the limitations of traditional data collection methods for estimating survival of juveniles, which are typically difficult to track and accurately count through time (Bloom et al. 2012, Riley and Conway 2020). By leveraging the dependency of ducklings on adults, and using drone-based technology, our method provides a less invasive alternative than marking individual offspring for studies focused on estimating juvenile survival. However, we are assuming that any juvenile no longer with its adult parent has not survived, which may not always be true where substantial adoption or brood mixing occur, such as in waterfowl. Street et al. (2022) present a modeling framework that accounts for adoption and brood mixing of unmarked juveniles amid imperfect detection.

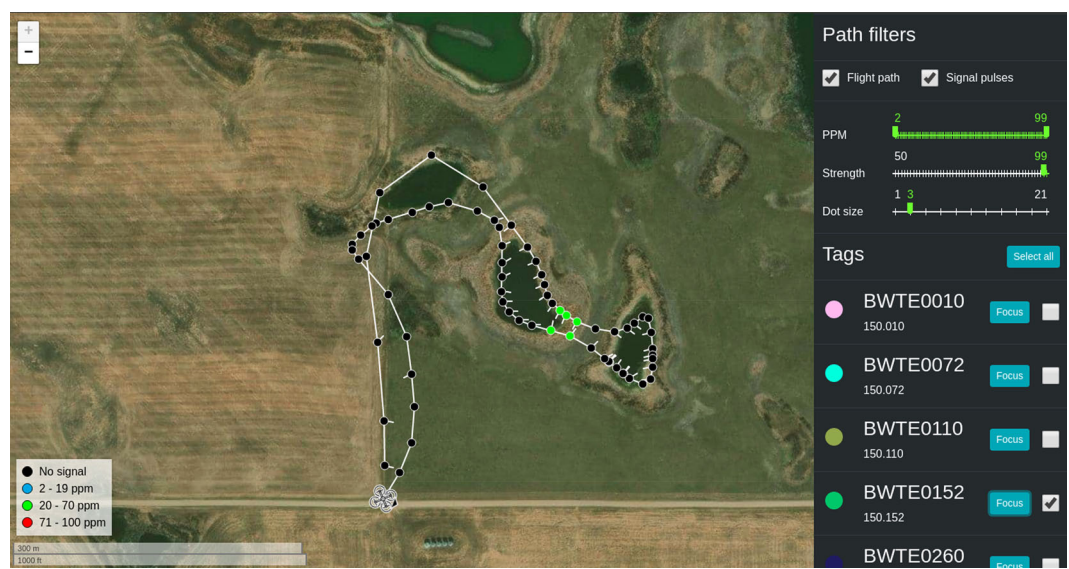


FIGURE 5 The VHF drone receives the maximum number of pulses per minute (PPM) emitted by the radiotransmitter. As the receiver gets closer to the transmitter, the stronger and more consistent the signal pulses are received. Once a signal is acquired for a specific frequency, the software program from Wildlife Drones® filters the data to identify the highest signal strength, based on the number of PPM detected, while filtering out background noise and irrelevant detections.



FIGURE 6 Triangulation techniques, using radio signal frequency cues, involved 3 or more rotations of the antennae. Bearings (green arrows) were estimated from each rotation point, and the intersections of these bearings created a mean estimated location (bold green point). The error ellipse (transparent green area) represents the uncertainty around this estimated location, with its size indicating the precision of the signal's triangulated position.

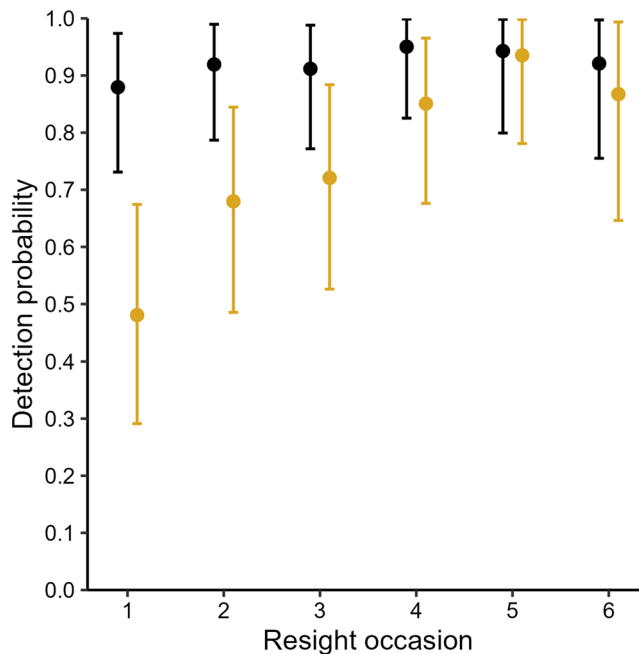


FIGURE 7 Detection probabilities of blue-winged teal broods across 6 resight occasions (2 occasions during the first week (3- and 7- day intervals), then weekly as brood got older, for a total of 35 days), using 2 different drone-based survey methods: VHF radiotelemetry (black points) and multispectral camera (yellow points). Error bars represent 95% credible intervals for each detection probability estimate.

Drone-based radiotelemetry systems offer researchers the ability to survey larger areas more efficiently, covering more ground and overcoming signal interferences that often hinder ground-based methods like handheld or vehicle-mounted systems. Saunders et al. (2022) found that drones achieved substantially higher survey coverage across various landscapes, from forested, mountainous areas to flat, undulating terrain, compared to on-foot tracking. In our prairie landscape, the VHF drone allowed us to locate multiple radio-marked females from a single location, usually within an hour. By reducing time spent locating radio-marked individuals, more time is available to collect ancillary information about the study species that should help interpret target parameters, such as behavior, forage, or vegetation information. However, system-specific research is needed to integrate drones in researchers' respective fields of study. In systems where focal species are not concentrated on the landscape, researchers should consider using triangulation techniques with the VHF drone to locate radio-marked individuals.

Current methods for obtaining counts of offspring include using spotting scopes, binoculars, or tripod-mounted cameras, as well as more invasive techniques such as beat-out or flush surveys (Amundson and Arnold 2010, Dahlgren et al. 2010, Lyons et al. 2020). However, given the potential for traditional methods like flushing to negatively affect juvenile survival (Street et al. 2022), researchers should consider the use of drones as a less invasive alternative to mitigate the risk of disturbing young during data collection. Drones can conduct aerial surveys and monitoring, which minimizes disturbance to the focal species (Ryckman et al. 2022). In our study, we observed minimal response to the drone in terms of flushing or evasive movements. At most, females and their ducklings were aware of the drone but would not change their behaviors (foraging, loafing) in response to it. By integrating drone technology into survey designs, researchers can enhance efficiency and uphold ethical standards, contributing to more sustainable study efforts.

Given our 6-interval sampling approach, the probability of detecting an unmarked duckling with the camera drone was $p = 0.48$ (95% CI: 0.293–0.673) during the initial survey and increased to near 1.0 as broods aged (Figure 7). Lower detection estimates of young broods were a result of brood behavior (Diem and Lu 1960, Ringelman and Flake 1980).

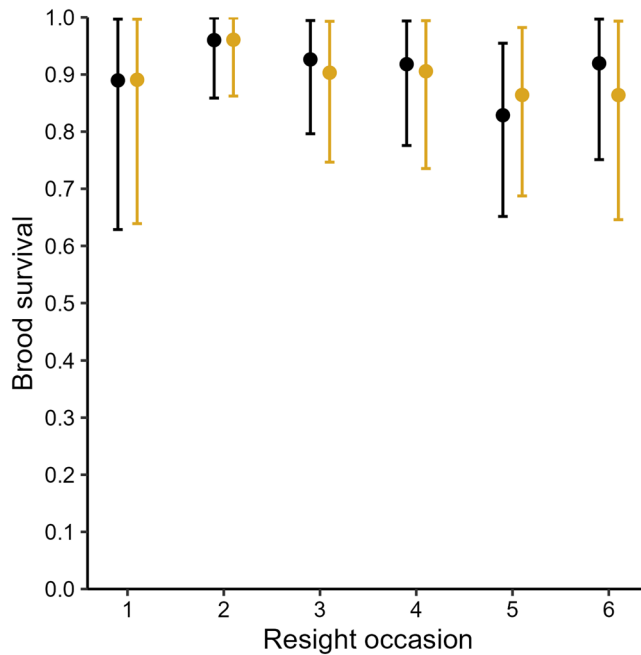


FIGURE 8 Survival probabilities of blue-winged teal broods across 6 resight occasions (2 occasions during the first week (3- and 7- day intervals), then weekly as brood got older, for a total of 35 days), using 2 different drone-based survey methods: VHF radiotelemetry (black points) and multispectral camera (yellow points). Error bars represent 95% credible intervals for each detection probability estimate.

Detection increased as broods got older, when females spent less time brooding their offspring in the emergent vegetation and juveniles became more active. Our imperfect detection of marked females with the camera drone, when using location cues from the VHF drone, demonstrates that the assumption of perfect detection of unmarked group members associated with radiomarked individuals in radiotelemetry studies may be violated (Williams et al. 2020). Researchers should develop appropriate survey designs with detection probabilities in mind to expressly state and minimize violations of this assumption (Pagano and Arnold 2009, Riley and Conway 2020, Riley et al. 2021).

Visibility bias, or non-detection, is a significant source of error in waterfowl brood and other wildlife survey techniques (Cowardin and Blohm 1992, Giudice 2001). Repeated, independent counts of the same area have been used to estimate detection probabilities and correct for visibility bias (Pagano and Arnold 2009, Bushaw et al. 2021). Repeated counts require assumptions about what constitutes a recapture of an unmarked brood (i.e., the same species, on the same wetland, with the same number of ducklings at the same age class as the first sighting; Pagano and Arnold 2009). Furthermore, replicate count modeling approaches rely on the assumption that all individuals are present and available for detection at least once during each survey (Pagano and Arnold 2009). Bushaw et al. (2021) used repeat-visit surveys to estimate detection probabilities of unmarked broods from both ground- and drone-based methods and found that not all broods were available to be detected using ground-based approaches, and that ground surveys missed approximately half of all broods observed with the drone. One result from our study is that the thermal/optical camera was unable to locate some broods (mean detection = 0.76) that were confirmed to be present on the wetland via radiotelemetry (mean detection = 0.92) providing an initial correction factor for the percentage of broods that successfully elude researchers using a thermal camera-based methodology. Bushaw et al. (2021) achieved a lower drone-based detection probability of 0.55 for unmarked broods because, as our work here has shown, not all broods are able to be sighted with the thermal camera, and because broods may have moved between wetlands and violated their unmarked brood recapture assumptions. In general, our hierarchical marking

system of adult females can be used to estimate and compare detection probabilities across different monitoring protocols used for waterfowl brood surveys.

The field sampling techniques described here are useful for studying wildlife with similar monitoring challenges, especially when integrated with new modeling frameworks (Schmidt et al. 2015, Williams et al. 2020, Street et al. 2022). For instance, birds that rear their broods in open habitats could be monitored more efficiently and with less disturbance compared to existing methods (Sharpe et al. 2009, Riley and Conway 2020, Riley et al. 2021). Drone-based approaches may be a suitable fill-in for the gap between ground-based and aircraft-based data collection methods when studying ungulate or carnivore species, which are difficult to detect and track due to their high mobility and location in remote areas (Adams et al. 2008, Grovenburg et al. 2013). Overall, our methodology has the potential to advance wildlife research and conservation efforts across a wide range of species and ecosystems, providing researchers accurate and efficient data collection methods for monitoring target parameters such as behavior, habitat use, and population dynamics.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ETHICS STATEMENT

Research was conducted under Saskatchewan Academic Research Permit no. 23AR008; Environment and Climate Change Canada banding permit no. 10965 A; scientific collecting permit no. SC-PR-2023-MB-SK-027. Radio-telemetry operations were conducted under Innovation, Science, and Economic Development Canada radio license no. 010993068-001. Drone operation was conducted under Transport Canada Special Flight Operations Certificate – Remotely Piloted Aircraft System permit no. 940008. All capture and handling methods were approved by Louisiana State University Institutional Animal Care and Use Committee Protocol no. A2022-26.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available upon request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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